RAJALAKSHMI ENGINEERING COLLEGE

Dhanushya M 251801054-2025-AIDS

D2

Started on	Thursday, 20 August 2026, 8:38 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 5:07 PM
Time taken	5 days 8 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int count=0;
    if(scanf("%d",&n)!=1)
    {
        return 1;

    }
    int i=1;count++;
    int s=1;count++;
    while(count++,s<=n){
        i++;count++;
        s+=i;count++;
    }
    printf("%d",count);
    return 0;

}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Dhanushya M 251801054-2025-AIDS

D2

Started on	Thursday, 20 August 2026, 8:56 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 5:13 PM
Time taken	5 days 8 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n)
{
    int count = 0;

    if(n == 1)
    {
        count++;          // if condition
        printf("*");
        count++;          // printf("*")
    }
    else
    {
        count++;          // if condition

        for(int i = 1; i <= n; i++)
        {
            count++;      // i <= n

            for(int j = 1; j <= n; j++)
            {
                count++;  // j <= n


                count++;  // first printf

```

```
                count++;  // second printf

                break;
            }

            count++;      // i++
        }

        count++;          // final i <= n
    }

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    func(n);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Dhanushya M 251801054-2025-AIDS

D2

Started on	Thursday, 17 September 2026, 8:07 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:23 AM
Time taken	15 mins 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, i;
    int count = 0;

    scanf("%d", &n);


    for (i = 1; i <= n; ++i)
    {
        count++;

        if (n % i == 0)
        {
            count++;
        }
        count++;
    }
    count++;
    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔

	Input	Expected	Got	
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Dhanushya M 251801054-2025-AIDS

D2

Started on	Thursday, 17 September 2026, 8:27 AM
State	Finished
Completed on	Thursday, 8 October 2026, 9:35 AM
Time taken	21 days 1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int c = 0;
    scanf("%d", &n);
    int i = n / 2;
    c++;
    for (; i < n; i++)
    {
        c++;
        int j = 1;
        for (; j < n; j = 2 * j)
        {
            c++;
            int k = 1;
            for (; k < n; k = k * 2)
            {
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d", c);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Dhanushya M 251801054-2025-AIDS

D2

Started on	Thursday, 17 September 2026, 9:05 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:09 AM
Time taken	3 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, rev = 0, remainder;
    int count = 0;

    scanf("%d", &n);

    count++;

    while (n != 0)
    {
        count++;
        remainder = n % 10;

        count++;
        rev = rev * 10 + remainder;

        count++;
        n /= 10;

        count++;
    }

    count++;
    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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